

Variation in albumin glycation rates in birds suggests resistance to relative hyperglycaemia rather than conformity to the pace of life syndrome hypothesis

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eLife Assessment

This **important** study uses extensive comparative analysis to examine the relationship between plasma glucose levels, albumin glycation levels, and diet and life history, within the framework of the "pace of life syndrome" hypothesis. The evidence that glucose is positively correlated with glycation levels and lifespan is **convincing** and, although there are some limitations related to data collection, they likely make the statistically significant findings more conservative. As the first extensive comparative analysis of glycation rates, life history, and glucose levels in birds, the study has the potential to be of interest to evolutionary ecologists and the aging research community more broadly.

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Abstract

The pace of life syndrome hypothesis (POLS) suggests that organisms' life history, physiological and behavioural traits should co-evolve. In this framework, how glycaemia (i.e., blood glucose levels) and its reaction with proteins and other compounds (i.e. glycation) covary with life history traits remain relatively under-investigated, despite the well documented consequences of glucose and glycation on ageing, and therefore potentially on life history evolution. Birds are particularly relevant in this context given that they have the

highest blood glucose levels within vertebrates and still higher mass-adjusted longevity when compared to organisms with similar physiology as mammals. We thus performed a comparative analysis on glucose and albumin glycation rates of 88 bird species from 22 orders, in relation to life history traits (body mass, clutch mass, maximum lifespan and developmental time) and diet. Glucose levels correlated positively with albumin glycation rates in a non-linear fashion, suggesting resistance to glycation in species with higher glucose levels. Plasma glucose levels decreased with increasing body mass but, contrary to what is predicted to the POLS hypothesis, glucose levels increased with maximum lifespan before reaching a plateau. Finally, terrestrial carnivores showed higher albumin glycation compared to omnivores despite not showing higher glucose, which we discuss may be related to additional factors as differential antioxidant levels or dietary composition in terms of fibres or polyunsaturated fatty acids. These results increase our knowledge about the diversity of glycaemia and glycation patterns across birds, pointing towards the existence of glycation resistance mechanisms within comparatively high glycaemic birds.

Introduction

The pace of life (POL) hypothesis [1–3] postulates that organisms behavioural/physiological characteristics and life histories have co-evolved in answer to specific ecological conditions forming pace of life syndromes (POLS) along a fast-slow continuum (but see [4] for the more recent consideration of other axes). According to the classical approach from life history theory, slower organisms would have, e.g. large body mass, late maturation, slow growth rate, small number of large offspring and high longevity, while faster ones representing the opposite trait values (see [5–8]) within this continuum. Pace of life hypothesis added physiology to this continuum, with studies testing its predictions focusing often on metabolic rate (i.e. higher or lower metabolic rate corresponding to faster or slower pace of life, respectively; see e.g. [9–12]). However, key energy substrates related to metabolic performance, such as glucose concentrations in tissues, have been largely overlooked (but see [13–16]). Notably, glucose, a central energy source for many organisms, plays a pivotal role in metabolism and there are some indications that its circulating blood or plasma levels correlate positively with metabolic rate per gram of mass at the interspecific level [13, 17], and with whole body metabolism at the intraspecific level ([18] for the case of non-diabetic humans).

Research focusing on plasma glucose becomes highly relevant as high glycaemia can entail costs that accelerate ageing. Along with other reducing sugars, glucose can react non-enzymatically with free amino groups of proteins, lipids or nucleic acids, a reaction known as glycation [19][20]. This reaction, after several molecular reorganizations, can lead to the formation of a plethora of molecules called Advanced Glycation End-products (AGEs) [21]. AGEs can form aggregates in tissues that are well known to contribute to several age-related diseases, such as diabetes (e.g., reviews by [22–25]). AGEs are also known to promote a proinflammatory state through their action on specific receptors called RAGEs [26–27].

Remarkably, birds show circulating glucose levels much higher than other vertebrate groups, on average almost twice as high as mammals [28]. These relatively high glucose levels might support the higher metabolic rates [29, 11] and body temperatures observed in birds compared to mammals [30–31]. In addition, elevated glycaemia is thought to be an adaptation to flight, providing birds with rapid access to an easily oxidizable fuel during intense bursts of aerobic exercise (see [32–33]), such as during take-off and short-term flapping flights [34–35]). However, the presence of high glycaemia in birds is also paradoxical, given their remarkable longevity compared to their mammalian counterparts — living up to three times longer than mammals of equivalent body mass [36, 11].

Several non-excluding hypotheses have been proposed to explain how birds apparently resist the pernicious effects of a high glycaemic state. One possibility is that birds might resist protein glycation through a lower exposure of lysine residues at the surface of their proteins [37]. Another hypothesis suggests that increased protein turnover may play a role [38–39]. Additionally, birds might benefit from more effective antioxidant defences [40–42] (although within Passeriformes, [16] shows no coevolution between glycaemia and antioxidant defences) or even the presence of “glycation scavengers” (i. e. molecules that bind glucose avoiding it to react with proteins) such as polyamines which circulate at high concentration [35]. Moreover, it is generally considered that RAGEs are not present in avian tissues, suggesting possible adaptive mechanisms that would allow birds to avoid the inflammatory consequences associated with RAGE activation [43–44]. However, a putative candidate for RAGE and a case of AGE-induced inflammation in birds were identified by Wein et al. [45], warranting further investigation on the occurrence of RAGEs in birds.

To date, although protein glycation levels have been measured in birds in several species, these results were mostly descriptive and concerned a small number of species [46–54]. Additionally, many of these studies employ commercial kits that were not specifically designed for avian species, making it challenging to interpret the results [53]. These limitations restrict our ability to draw general conclusions from these studies. Here, we performed a comparative analysis on primary data (for glycation and most of glucose values, see methods and ESM6) from 88 bird species belonging to 22 orders (see ESM6) and assessed whether and how bird glycaemia and glycation rates are linked to ecological and life-history traits (including body mass). While doing this, we also checked whether glycation levels are influenced by circulating plasma glucose concentrations. Finally, as diet can influence glycaemia (see e.g. [15, 55–56] but [14] found no significant effects) and glycation, we included it in our analyses. We hypothesized that carnivorous birds would exhibit higher glycaemia and glycation levels than omnivorous or herbivorous species, as diets with low-carbohydrates, high-proteins and high-fat are associated with comparatively high glycaemia and reduced insulin sensitivity in some vertebrates (e.g.; [57–59]). Accordingly, a recent comparative study in birds showed higher blood glucose levels in carnivorous species [56]. This may be attributed to high levels of constitutive gluconeogenesis, which has been confirmed in certain raptors [60–61]. Conversely, bird species with high sugar intake, such as frugivores or nectarivores, are expected to exhibit high glycaemia, as observed in hummingbirds [49] and Passeriformes with such diet [15] (although the opposite was found in [56]). In line with POLS hypothesis, we also hypothesized that species with a slower POL should exhibit a lower glycaemia and higher glycation resistance (glycation levels for a given glycaemia) compared to species with a faster POL. Hence, we included the following four parameters in our analyses of glycaemia and glycation levels: body mass, maximum lifespan, clutch mass and developmental time. Body mass is one of the main factors underlying life history, with higher body mass associated with “slower” strategies and vice versa [62]. Previous studies have reported a negative relationship between body mass and glycaemia [63, 34, 13–14, 56] (but see [49, 64] with non-significant trends in birds, and [15] only for temperate species within Passeriformes). Therefore, we predicted that bird species that live longer, develop more slowly and invest less per reproductive event (see ESM6 for further justification of the chosen variables) should show lower plasma glucose levels and albumin glycation rates (after controlling for body mass, phylogeny, diet and glucose in the case of glycation; see methods).

Materials and methods

Species and sample collection

A total of 484 individuals from 88 species measured were included for this study (see [Table ESM6.1](#)). A detailed list with the provenance of the samples, including zoos, a laboratory population, and both designated captures and samples provided by collaborators from wild populations, is provided in ESM3, with a textual description in ESM6.

Blood samples were collected from the brachial or tarsal vein, or from the feet in the case of swifts, using heparin-lithium capillaries or Microvette® (Sarstedt). As samples were collected mostly by different collaborators, handling times were not always recorded and could not be adjusted for, potentially rendering the results more conservative (for a more detailed discussion on potential stress effects on glucose, see ESM6). Samples were centrifuged at 4°C, 3500 g for 10 min and plasma was aliquoted when a large volume was available. Subsequently, they were transported on dry ice to the IPHC (Institut Pluridisciplinaire Hubert Curien) in Strasbourg and stored at -80° C until analysis. Glycaemia was measured in the laboratory on the remaining plasma after taking an aliquot for glycation assessment, using a Contour Plus® glucometer (Ascensia diabetes solutions, Basel, Switzerland) and expressed in mg/dL. These of point-of-care devices have previously been assessed for its usage in birds [[65](#)–[66](#)], with several examples of its usage in recent literature related to the topic presented here (e.g. [[14](#), [67](#)–[69](#)]). We also performed an assay with this particular brand on a subset of 46 samples from this study, coming from nine species distributed across the whole range of glucose values (three species with “low” values, three with “medium” and three with “high”, from both captive and wild populations, with five individuals per species, except one including six), confirming a positive linear correlation ($R^2_{\text{marginal}}=0.66$; $R^2_{\text{conditional}}=0.84$, for a model including the species as random factor) of this device with Randox GLUC-PAP colourimetry kits (P-value<0.001, unpublished data). Due to technical issues, related to the incapability of the device to determine certain glucose values (not because of the glucose concentration, but perhaps the particular composition of the plasma samples from certain species), we could not determine glycaemia values of 95 individuals of those that we sampled and in which glycation levels were assessed (belonging to 40 species coming from different sources). For these species, if not a single individual had a glucose measurement (which was the case in 13 species), we obtained mean plasma glucose values reported for the species from the ZIMS database from Species360 (Species360 Zoological Information Management System (ZIMS) (2023), [zims.Species360.org](#)). This database provides plasma glucose data measured by colorimetric glucose kits on zoo specimens not necessarily corresponding to those in which we measured glycation values. The sample sizes for both glucose and glycation measurements, either from our measured individuals or from the ones from ZIMS, are reported on the ESM5.

Glycation levels

Glycation levels for each individual were determined using liquid chromatography coupled to mass spectrometry (LC-MS), which is considered the gold standard for the assessment of protein glycation levels (see e.g. [[70](#)]) and has previously been used for birds [[52](#)–[53](#), [71](#)]. Given the relatively high time intrinsically taken by the employed methodology for analysing all the samples, linked to constraints in the access to the mass spectrometry devices, the whole set of samples of this study were analysed across several instances within a total timespan of less than 2 years (2021-2023). Only one sample from each individual was measured, given logistic limitations on the total number of samples that could be processed. Briefly, 3 µl of plasma were diluted with 22 µl of distilled water containing 0.1% of formic acid, followed by injection of 5 µl into the system. The glycation values used in the analyses represent the total percentage of glycated albumin, obtained by adding the percentages of singly and doubly glycated albumin. These percentages were calculated by dividing the areas of the peaks corresponding to each glycated molecule form (albumin plus one or two glucose) by the total albumin peak area (sum of the areas of glycated plus non-glycated molecules) observed in the spectrograms obtained from the mass spectrometry outcomes. These spectrograms represent the different intensities of signal for the components

differentiated in the sample by their TOF (Time Of Flight), which depends on the mass to charge ratio (m/z) of the ionized molecules. More detailed information regarding functioning of the method and data processing can be found in [53]. In cases where albumin glycation values dropped below the limit of detection, resulting in missing data, these individuals (10 individuals from 4 species) were excluded from the statistical analyses, as outlined below.

Data on ecology and life-history

Diets of individual species were extracted from the AVONET dataset ([72]), coming from [73], and those adapted from [74]) and life history traits from the Amniote Database [75]. After determining that AVONET diet classifications did not align with our research needs, minor changes were made after consulting the original Wilman et al. [74] database (see ESM6). For missing data on life history traits after this stage, we extracted values, in order of preference, from the AnAge database [76] or the Birds of the World encyclopaedia [77], by calculating mean values if an interval was given, and then averaging if male and female intervals were provided separately. For European species, maximum lifespan records have always been checked against the most recent Euring database [78], and from *Chionis minor* and *Eudiptes chrysolophus* (not available from the previous sources) they were extracted from the Centre d'Etudes Biologiques de Chizé, Centre National de la Recherche Scientifique (CNRS-CEBC) database. Several species from the zoo had maximum lifespan values available on the ZIMs from Species 360 (Species360 Zoological Information Management System (ZIMS) (2023), zims.Species360.org), so these were also compared with the data we had from other sources.

In the case of variables (other than maximum lifespan) for which two different sources provided different records, an average value was calculated. For the maximum lifespan value, available sources were cross-checked and the highest value was always used. In the absence of satisfactory support from another source, values for maximum lifespan indicated as being anecdotal and of poor quality have been excluded from the analyses. A table with the adequate citations for each value is provided as part of the online available data (see Data accessibility section). When the source is AVONET [72], this database is cited, and when it is Amniote Database [75], we cite the sources provided by them, so the references can be checked in [75]. A list with all the additional references not coming from any of these databases or not provided by their authors (and that are not already in the main text) is given as ESM4.

Statistical analyses

All analyses were performed in R v.4.3.2 [79]. Alpha (α) ≤ 0.05 was reported as significant, and $0.05 < \alpha \leq 0.1$ as trends. General Linear Mixed Models with a Bayesian approach (*MCMCglmm* function in R; [80]) were performed. All models were run with 6×10^6 iterations, with a thinning interval of 100 and a burn-in of 1000. The models were simplified by eliminating quadratic terms where they were not significant and selecting models with lower AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion). Gaussian distribution of the response variables was assumed (after $\log_{10}$ conversion for glucose; see ESM6 for further discussions). The priors were established assuming variance proportions of 0.2 for the G matrix and 0.8 for the R matrix, with $\nu=2$. Less informative priors with equal variance proportions for each partition (residual and random) gave similar results. Lower ν values (0.1, 0.2, 0.002) were also tested, without success (i.e. simulations aborted before the established number of iterations). For the phylogenetically controlled analyses, consensus trees (each with the species included in the model) were obtained by using the *consensus.edges* function (from the *phytools* package from R [81]) from a total of 10,000 trees downloaded from *Birdtree.org* [82] (option “Hackett All species” [83]). For such purpose, a list of species with the names adapted to the synonyms available in such website was used (see ESM5), including the change of *Leucocarbo verrucosus* for *Phalacrocorax campbelli*, as the former was not available and it was the only species from the order Suliformes in our dataset, so that neither the position in the tree nor the branch length would be affected by this change. The consensus trees were included as *pedigree* in the models. We

performed models with either glucose or glycation as dependent variables and with the diet, body mass and life history traits as predicting variables (and glucose in the models with glycation as a response to assess glycation resistance; see introduction). Generalized Variance Inflation Factors (GVIFs) were calculated for all the models with more than one predictor variable to assess the collinearity of them, as it may be expected for life history traits (see results on ESM1), considering values above 1.6 as slightly concerning, above 2.2 as moderately concerning and above 3.2 as severely concerning (i.e. indicating high collinearity). Models testing the effects of age and sex on glucose and glycation levels and the number of exposed lysine residues on glycation were also carried out. Finally, we performed models on glucose and glycation values controlling for the taxonomic orders included in the dataset (see ESM5). The effects of phylogeny on all models was determined by calculating the ratio of the variance estimated for the “animal” variable, representing the *pedigree* (see ESM6) by the total variance (all random factors plus *units*). A thorough description of all of the models, including transformations of the variables and other details is given on the ESM6.

Results

Plasma glucose and albumin glycation variability across the birds’ tree

Plasma glucose and albumin glycation values varied considerably across species (**Figure 1** [↗](#)). Considerable within species repeatability (see ESM6) was observed (Glucose: $R = 0.716$, $SE = 0.042$, $CI_{95} = [0.619, 0.785]$, $P\text{-value} < 0.0001$; Glycation: $R = 0.703$, $SE = 0.042$, $CI_{95} = [0.603, 0.768]$, $P\text{-value} < 0.0001$).

We found significant differences between some orders for both glucose and albumin glycation. ESM1 shows the raw outcomes of the models, showing such significant differences to the intercept and the Credible Intervals to perform the pairwise comparisons across all groups. To explore some of the details, we can see for example how Apodiformes showed the highest average glucose values (species average model: Estimated mean=351.8 mg/dL, $CI_{95}[238.9, 514.2]$; model with individuals: Estimated mean=349.8 mg/dL, $CI_{95}[251.8, 476.7]$, $n=39$ individuals, 1 species), followed by Passeriformes (species average model: Estimated mean=349.2 mg/dL, $CI_{95}[279.6, 438.6]$; model with individuals: Estimated mean=337.6 mg/dL, $CI_{95}[274.5, 414.5]$, $n=84$ individuals, 8 species), while the Suliformes (species average model: Estimated mean=172 mg/dL, $CI_{95}[117.6, 252.9]$; model with individuals: Estimated mean=169.5 mg/dL, $CI_{95}[120.9, 239.7]$, $n=5$ individuals, 1 species), Rheiformes (species average model: Estimated mean=173.7 mg/dL, $CI_{95}[118, 254.7]$; model with individuals: Estimated mean=172 mg/dL, $CI_{95}[123.7, 239.8]$, $n=9$ individuals, 1 species) and Phoenicopteriformes (species average model: Estimated mean=176 mg/dL, $CI_{95}[130, 241.1]$; model with individuals: Estimated mean=141.2 mg/dL, $CI_{95}[100.4, 198.2]$, $n=5$ individuals, 1 species) show the lowest average values (for the species average models, the pattern is Suliformes<Rheiformes<Phoenicopteriformes, while for the individual models is Phoenicopteriformes<Suliformes<Rheiformes).

For glycation, Strigiformes (species average model: Estimated mean=25.6 %, $CI_{95}[20.4, 31.1]$; model with individuals: Estimated mean=25.3 %, $CI_{95}[19.1, 31.5]$, $n=7$ individuals, 2 species), Apodiformes (species average model: Estimated mean=25.5 %, $CI_{95}[18.6, 32.4]$; model with individuals: Estimated mean=24.8 %, $CI_{95}[17.9, 32]$, $n=40$ individuals, 1 species) and Coraciiformes (species average model: Estimated mean=24.5 %, $CI_{95}[17.9, 31.6]$; model with individuals: Estimated mean=23.9 %, $CI_{95}[15.8, 32]$, $n=2$ individuals, 1 species), all terrestrial carnivores as by our sampled species (see discussion below), had the highest average.

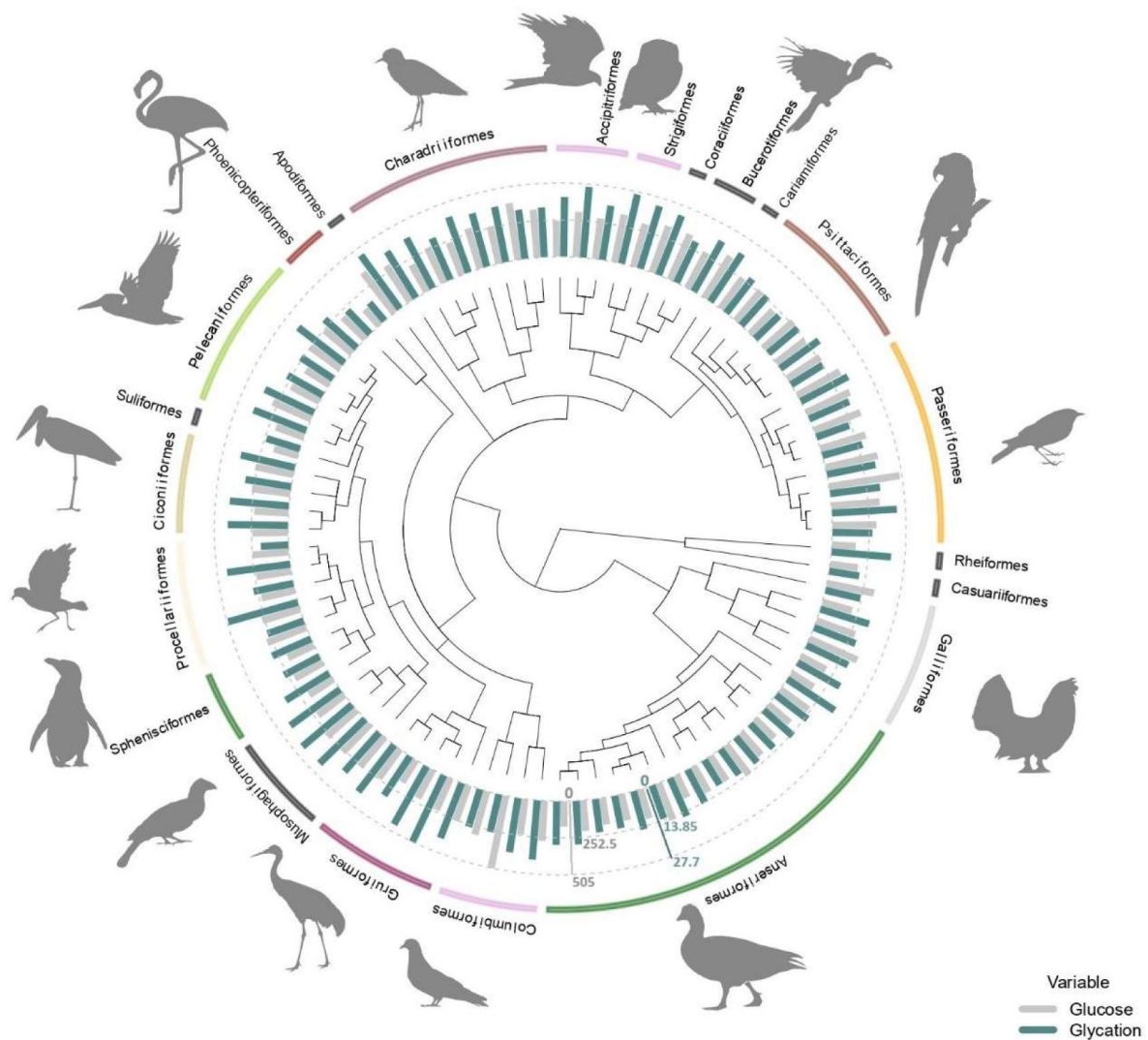


Figure 1.

Average plasma glucose values in mg/dL (in grey) and average albumin glycation rate as a percentage of total albumin (in blue) from all the species used in this study (some of them with glucose values coming from ZIMs database; see methods) with the orders they belong to. Glucose and glycation values are standardized in order to be compared, with the dotted lines representing half the maximum and maximum values for each variable (as indicated by the axes in their corresponding colours), from inside out. Tree from a consensus on 10,000 trees obtained from "Hackett All species" on Birdtree.org, including 88 species from 22 orders (see methods).

On the other hand, Casuariiformes (species average model: Estimated mean=10.8 %, CI₉₅[3.7, 18.1]; model with individuals: Estimated mean=11.4 %, CI₉₅[2.7, 20.6], n=1 individuals, 1 species), Phoenicopteriformes (species average model: Estimated mean=11.2 %, CI₉₅[5.6, 16.7]; model with individuals: Estimated mean=10.5 %, CI₉₅[4.6, 16.3], n=22 individuals, 2 species) and Suliformes (species average model: Estimated mean=13.8 %, CI₉₅[7, 20.7]; model with individuals: Estimated mean=13.2 %, CI₉₅[5.6, 20.5], n=5 individuals, 1 species) had the lowest average levels for the species average models, the pattern is Casuariiformes<Phoenicopteriformes<Suliformes, while for the individual models is Phoenicopteriformes<Casuariiformes<Suliformes). Graphs with raw data on species average and individual glucose and glycation values by order are shown on ESM2 (Figure ESM2.3 [↗](#)). Estimates of phylogeny effects on the residuals of the models on glucose and glycation values differ if we consider the models with intraspecific variability or without it, but not so much between the models with and without life history traits (within the previous categories). In the case of species averages, the effect of the tree on residual glucose variation is lower and the estimation is less precise than for residual glycation variation, while for the models considering individual values, it is glycation residuals what shows lower levels of tree-related variance than glucose residuals (see ESM1).

Bird glycemia appears related to body mass and maximum lifespan

After controlling for intraspecific variation in our analyses, we found that variation in glucose levels were significantly explained by variations in body mass and both the linear and quadratic component of residual maximum lifespan (i.e. mass and phylogeny-adjusted maximum lifespan), but not by variations in diet (Figure 4.A [↗](#) with predictions from the model without life history traits and Figure ESM2.2.A [↗](#) with raw individual data from the same dataset), clutch mass and developmental time (see Table 1 [↗](#) for the model including life-history trait variables). Heavier species had lower glucose levels (see Figure 2.A [↗](#), drawn with the estimates from the model without life history traits, which includes more species). Glucose levels increase with increasing mass-adjusted lifespan until reaching a plateau (Figure 2.B [↗](#)). Models that did not consider intraspecific variability show no significant effect of any of the aforementioned variables on glucose levels (see ESM1). The provenance of the samples (wild versus captive) only showed a trend to a higher glucose levels in the samples from captive individuals in the model without life history traits (Estimate = 0.058; CI₉₅[-0.008, 0.125]; P-value=0.083).

Bird albumin glycation is related to glycemia and diet

After controlling for intraspecific variability, we found that diet was affecting variation in albumin glycation rates, with terrestrial carnivorous species having higher glycation levels than omnivorous species (Table 2 [↗](#) for complete model; see Figure 4.B [↗](#) with predictions of the model without life history traits and Figure ESM2.2.B [↗](#) with raw individual data from the same dataset). However, for the models for species averages, there was only a trend on this pattern in the one including life history traits, and no effect in the other (see ESM1). The relation between glycation and glucose levels was positive and significant in all but the model that included life history traits but not intraspecific variation (see Table 2 [↗](#) for the outcome of the model with individual values and life history traits and ESM1 for the rest; see Figure 3 [↗](#) with estimates from the model including individual variation but no life history traits, as it contains more species and the estimates are similar). Given the logarithmic relationship between glycation and glucose (see ESM6), the slope lower than one (see Table 2 [↗](#)) implies that birds with higher glucose levels have relatively lower albumin glycation rates for their glucose, fact that we would be referring to as higher glycation resistance.

Additional analyses looking at the number of exposed lysines in the albumin sequence of a species show no effect of this variable on albumin glycation rates (α =10.39: CI₉₅ [-3.713, 24.993], β =0.246: CI₉₅[-0.144, 0.629], P_{MCMC}=0.196; ESM1).

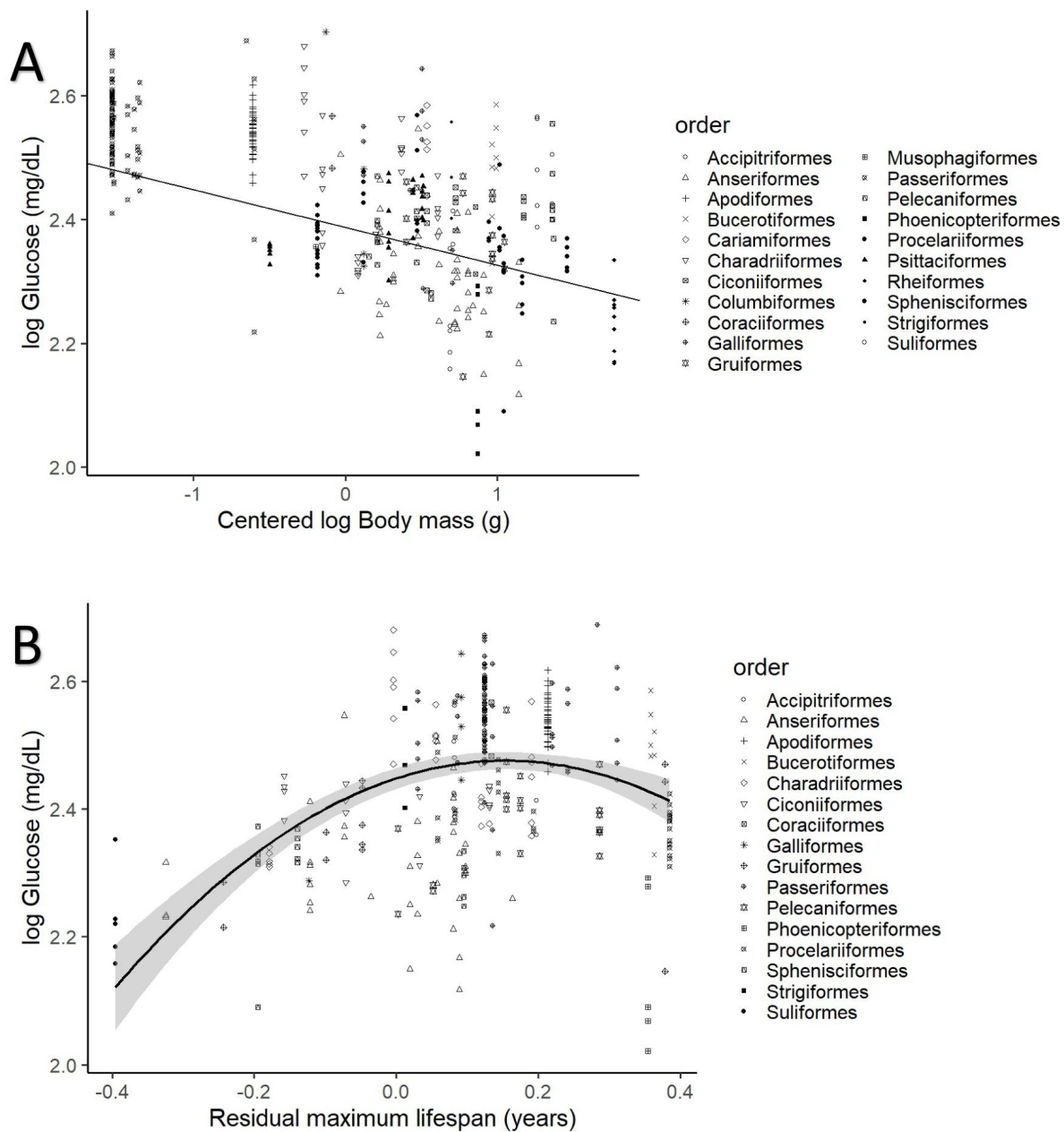


Figure 2.

Plasma glucose levels (in mg/dL) variation as a function of **A** species mean centred body mass and **B** residual maximum lifespan. Both glucose and body mass are log transformed. Maximum lifespan (in years) is given as the residues of a pGLS model with body mass (in grams), both $\log_{10}$ transformed, so the effects of body mass on longevity are factored out (see ESM6). Different bird orders, are indicated by symbols, as specified on the legends at the right side of the graphs. **Figure 2.A** uses the values and estimates from the glucose model without life history traits ($n=389$ individuals from 75 species), while **Figure 2.B** uses only the data points employed on the complete model ($n=326$ individuals of 58 species).

	Estimates	Lower 95% CI	Upper 95% CI	Sampling effort	pMCMC
Intercept (Omnivore)	2.387	2.268	2.5	59900	<0.001
Diet: Carnivore terrestrial	0.02	-0.056	0.099	59900	0.592
Diet: Aquatic predator	0.034	-0.047	0.112	59900	0.393
Diet: Herbivore	-0.053	-0.177	0.07	59900	0.393
Diet: Frugivore/granivore	-0.055	-0.239	0.135	59900	0.558
Centred Log ₁₀ Body mass	-0.061	-0.106	-0.015	59900	0.009
Maximum lifespan	0.107	-0.035	0.253	59900	0.142
Maximum lifespan ²	-0.616	-1.166	-0.095	59900	0.026
Clutch mass	-0.095	-0.265	0.069	59900	0.258
Developmental time	0.011	-0.185	0.212	59900	0.916
Provenance: Captive	0.034	-0.039	0.106	59900	0.346

Table 1.

Final glucose model with diet, body mass, life history traits and sample provenance (wild versus captive; see ESM6) as explanatory variables, including the significant quadratic effect of maximum lifespan. Posterior means, CI₉₅ and pMCMC from a phylogenetic MCMC GLMM model including n=326 individuals of 58 species. Both glucose and body mass are log₁₀ transformed and life history traits are residuals from a pGLS model of log₁₀ body mass and log₁₀ of the trait in question (see ESM6). Body mass was also centred to better explain the intercept, as 0 body mass would make no biological sense. The intercept corresponds to the Omnivore diet, being used as the reference as it is considered the most diverse and “neutral” group for this purpose. Significant predictors are indicated in bold.

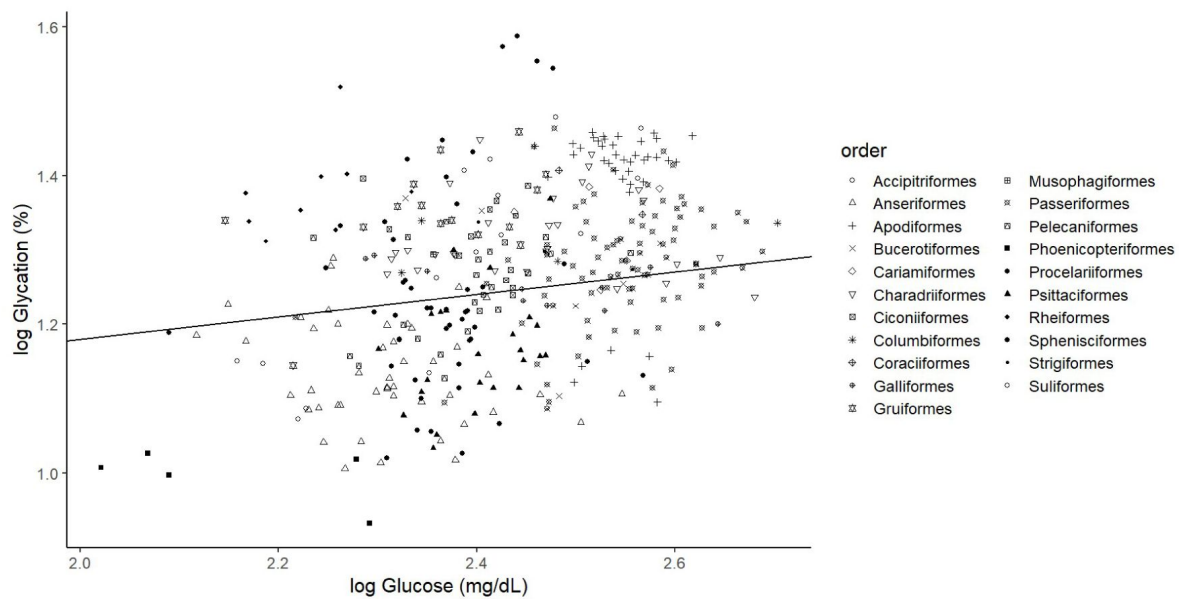


Figure 3.

Albumin glycation rates (as a percentage of total albumin, $\log_{10}$ transformed) variation as a function of individual plasma glucose values (mg/dL, $\log_{10}$ transformed). Different bird orders are indicated by symbols, as specified on the legends at the right side of the graphs. The values and estimates used are from the glycation model without life history traits ($n=379$ individuals from 75 species).

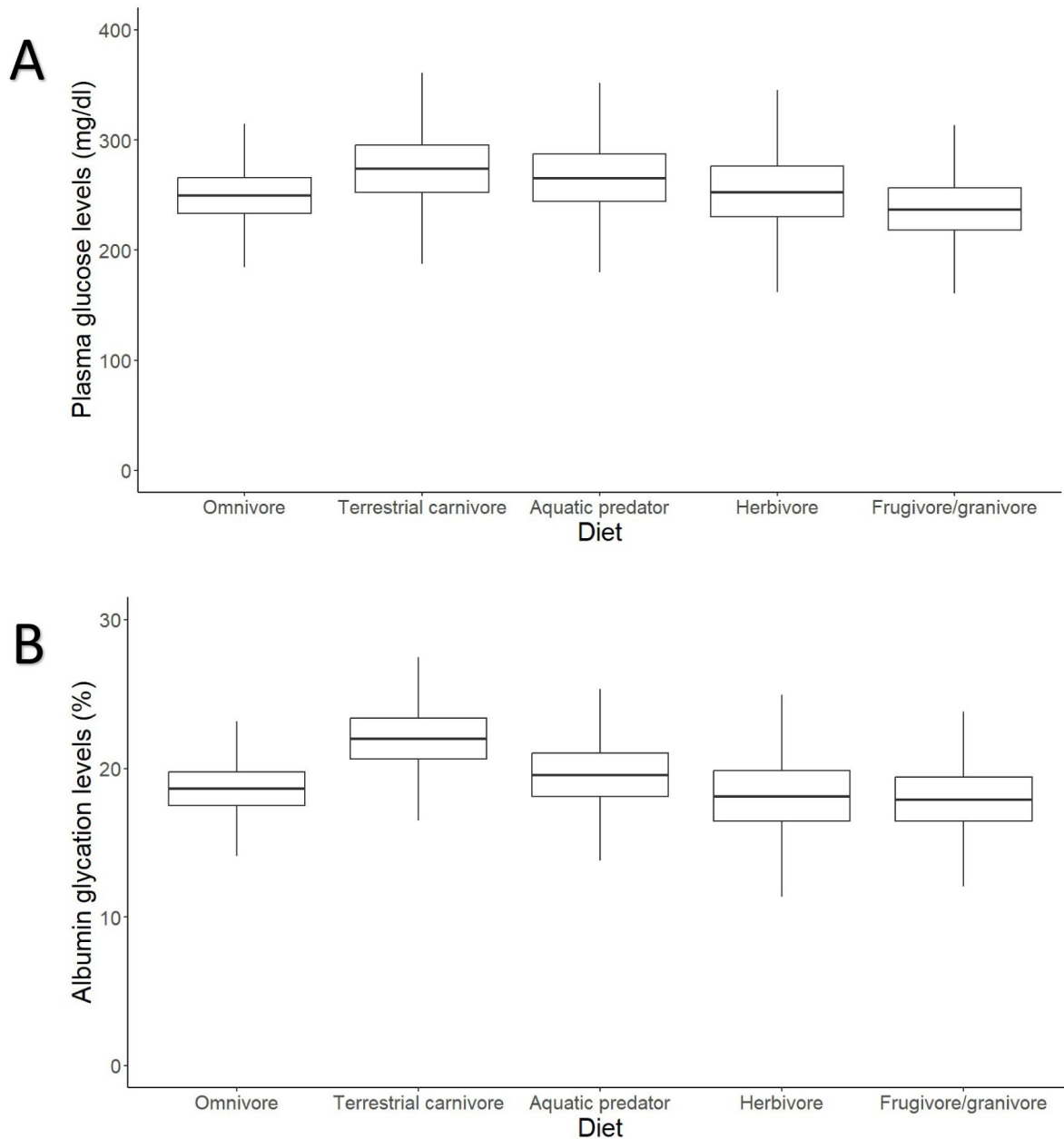


Figure 4.

Outcomes of the models on individual data on effects of diet on **(A)** plasma glucose levels and **(B)** albumin glycation in birds. Glucose levels are given in mg/dL, while glycation levels are a percentage of total plasma albumin which is found to be glycated. Terrestrial carnivores showed significantly higher glycation levels than omnivores (Estimate=21.62 %, CI₉₅[18, 25.95], $p_{MCMC}=0.049$). Models without life history traits, including more individuals, are represented, but the models with life history traits do not show differences in their qualitative predictions (i.e. higher albumin glycation in terrestrial carnivores than in omnivores; see ESM1).

	Estimates	Lower 95% CI	Upper 95% CI	Sampling effort	pMCMC
Intercept	1.232	1.112	1.351	59900	<0.001
Diet: Carnivore terrestrial	0.101	0.017	0.187	59900	0.021
Diet: Aquatic predator	0.027	-0.062	0.118	59900	0.549
Diet: Herbivore	-0.019	-0.161	0.119	59936	0.781
Diet: Frugivore/granivore	0.095	-0.098	0.292	59900	0.329
Centred log ₁₀ Body mass	0.004	-0.043	0.05	58765	0.876
Log ₁₀ Glucose	0.137	0.012	0.255	59900	0.027
Maximum lifespan	0.037	-0.122	0.195	59043	0.648
Clutch mass	0.151	-0.03	0.346	59900	0.114
Developmental time	0.04	-0.188	0.266	59303	0.725

Table 2.

Final glycation model with diet, body mass, glucose and life history traits as explanatory variables. Posterior means, CI₉₅ and pMCMC from a phylogenetic MCMC GLMM model including n=316 individuals of 58 species. Glycation, glucose and body mass are log₁₀ transformed and life history traits are residuals from a linear model of log₁₀ body mass and log₁₀ of the trait in question (see ESM6). Body mass and glucose were also centred to better explain the intercept. The intercept corresponds to the Omnivore diet, being used as the reference as it is considered the most diverse and “neutral” group for this purpose. Significant predictors are indicated in bold and the credible intervals are considered for making pairwise comparisons between the groups.

We observed no significant effect of age relative to maximum lifespan nor sex on either glycaemia or glycation (see ESM1). In some models, GVIF for body mass and/or clutch mass were higher than 1.6, and in one case body mass is slightly above 2.2 (2.26, see ESM1). This indicates that there may be moderate collinearity for this variables, but the fact that we used life-history trait covariates that are excluding body mass associated variation, this small effects remain mysterious, but nevertheless probably not worrying.

Discussion

Plasma glucose level and albumin glycation rate (co)variation patterns suggests resistance mechanisms

Our findings show that glucose levels vary widely among different bird groups. Interspecific differences are partly explained by the allometric relationship of glycaemia with body mass (**Figure 2.A**), which has already been reported in previous studies [56, 63, 34, 13, 14]. Indeed, Passeriformes, Apodiformes and some Columbiformes (e.g. *Nesoenas mayeri* holds the highest value in our dataset) are found at the higher end of the glucose level continuum, in accordance with their relatively small body mass and powered flight, while groups of larger birds such as Phoenicopteriformes, Anseriformes, Rheiformes and Suliformes tend to show low glycaemia levels. This pattern is similar for glycation, with some groups of large birds (such as Phoenicopteriformes, Anseriformes and Suliformes) showing the lowest levels of glycated albumin, while small birds (such as Apodiformes) had the highest values. Nevertheless, glycation remains high in some birds in relation to their glucose levels, as in Rheiformes, or low as in Psittaciformes or Passeriformes. The case of Procellariiformes, which are typically long-lived birds, is particularly striking, with some species exhibiting some of the highest glycation levels (*Calonectris diomedea* and *Macronectes giganteus*) and others some of the lowest (*Procellaria aequinoctialis*). This suggests that, if birds are protected against the deleterious effects of high glycaemia, certain species may have evolved mechanisms to efficiently prevent proteins to be glycated at high rates while others may have evolved mechanisms to resist the consequences of protein glycation. We should also bear in mind that some taxonomic groups may be underrepresented in our study, or biases in species selection due to availability contingencies (e.g. species common in zoos or present in European countries) may exist, so further studies should target this underrepresented groups in order to confirm our predictions.

As expected for a non-enzymatic reaction, just by the law of mass action, bird albumin glycation rates increase with plasma glucose concentration. However, the logarithmic nature of the relationship, and the fact that the slope is lower than one, suggest that species with higher plasma glucose levels exhibit relatively greater resistance to glycation. This finding aligns with previous research indicating that *in vitro* glycation levels of chicken albumin increase at a slower rate than those of human albumin when glucose concentration is elevated [83]. Moreover, these levels are consistently lower than those of bovine serum albumin regardless of glucose concentration and exposition time [37]. As discussed in previous studies comparing chicken to bovine serum albumin [37], or zebra finch to human haemoglobin [53], the lower glycation rates observed in bird proteins may result from a lower number of glycatable amino acids (e.g. lysines) in their sequence and/or their lesser exposure at the protein surface.

Our analyses do not succeed in indicating a significant positive relationship between average glycation levels and the number of glycatable lysines in the albumin sequence. This may be attributed to the limited number of species employed or the weak variation in the number of glycatable lysine residues, which are mostly ranging from 33 to 39. An interesting exception are the 18 glycatable lysine residues of flamingos (*Phoenicopiterus ruber*), which also shows very low

glycation levels (mean=10.1%). However, the exceptions of 44 in zebra finches and 20 in godwits (*Limosa lapponica*, used in place of *Limosa limosa*) are nevertheless associated with very similar average glycation levels of 20.2% and 20.7% respectively.

Plasma glucose relates with longevity and may be influenced by reproductive strategies

Our results were only in minor agreement with predictions from POLS theory: it holds for body mass, but not for the other life history variables tested. The only previous comparative study of glycaemia and life history traits to our knowledge [14] shows no relationship with mass-adjusted maximum lifespan in passerines. However, our study over 88 bird species on 22 orders revealed an increase in glucose with mass-adjusted longevity up to a plateau (see **Figure 2.B**). Thus, the relationship between glucose and maximum lifespan may depend on differences between bird orders or be tied to specific species particularities not explored in our study. Such species particularities might involve additional undetermined ecological factors that modify the relationship of glycaemia with longevity. Further exploration of glucose metabolism in relation with lifestyle will bring further light on species-specific life-history adaptations concerning glucose. For example, the species with the lower mass-adjusted maximum lifespan here was a cormorant (*Leucocarbo verrucosus*), which have quite low glucose values for birds.

Regarding reproductive investment (i.e. clutch mass), our results show no relationship with glycation (see **Table 2**), while previous studies reported positive relationships with glycaemia in passerines [14, 83]. Interestingly, most of the species with high clutch mass included in our study belong to the Anseriformes (**Figure ESM2.1**). While these species exhibit very low glycaemia and albumin glycation rates, they are also characterized by a particular reproductive strategy compared to passerines, for whom clutch mass does not imply the same in terms of parental investment. For instance, Anseriformes, unlike Passeriformes, are precocial and their offspring are not very dependent on parental care. Furthermore, they are capital breeders, accumulating the energetic resources required for reproduction in advance rather than collecting them during reproduction. These species typically have large clutch sizes and incubation is usually carried out by females, who store internal fat resources to support this main part of the parental investment. In addition, ducklings are born highly developed, reducing the amount of care required post-hatching (see e.g. [84–85]). Consequently, their dependence on glucose as a rapid energy source for reproduction may be lower, with investment in this activity likely more closely linked to lipid accumulation. This could explain why we did not detect previously reported effects of clutch mass on glucose levels.

Terrestrial carnivores show a paradoxically increased albumin glycation rate without increased plasma glucose levels

Contrary to our expectation of finding differences across dietary groups, plasma glucose did not significantly vary with species diet. This aligns with results previously reported by [14] for Passeriformes or by [55] for 160 species of vertebrates among which 48 were birds, but not with [15], that showed glycaemia to be increased with the proportion of fruits/seeds in the diet, or with [56], that showed higher (mass-adjusted) glucose levels for terrestrial carnivores and insectivores (those included in our terrestrial carnivores' category) and lower for frugivorous/nectarivorous. On the other hand, intraspecific data indicates that changes in diet composition hardly affects glycaemia in birds [86]. These studies suggest that glycaemia is tightly regulated independently of dietary composition within species, while it probably varies across species depending on the diet they are adapted to, although depending on the bird groups included in the analyses and the way of assessing it.

Although terrestrial carnivore species did not have significantly higher glycaemia levels in our study, they nonetheless demonstrated significantly higher albumin glycation rates, suggesting a potential susceptibility to protein glycation. Besides unique structural features, this phenomenon could be due to lower albumin turnover rates in terrestrial carnivores. Studies in humans have linked higher oxidative damage to albumin with lower turnover rates (reviewed in [87]), which may extend to other species and post-translational modifications of albumin, such as bird's albumin glycation. However, the contrary outcome would be anticipated if protein intake were high as seen in carnivorous species [87–88].

Interestingly, the contrast between terrestrial and aquatic predators, which did not show such high glycation rates, suggests the beneficial influence of a substantial difference in food composition or the pre-eminence of another ecological factor, yet to be determined, which may explain lower glycation levels in aquatic predators. In terms of food composition, the proportion of essential n-3 PUFA (polyunsaturated fatty acids), particularly long-chain ones such as DHA and EPA compared with n-6 PUFA and short-chain n-3 PUFA, is different in aquatic and terrestrial environments [89] and therefore in the diet of predators in each environment (e.g. [90]). For instance, low n-6 PUFA/n3-PUFA ratio have been shown to decrease insulin levels and improve insulin resistance in humans [91]. On the other hand, while some studies report that an increase in dietary n3-PUFA reduces the levels of glycated haemoglobin (HbA1c) in humans, this effect was not detected in many other studies and therefore remains inconclusive (reviewed in [92]). The detailed study of the fat composition of the diets of terrestrial and aquatic predators, in particular the ratio between different types of PUFA, merits more attention. In addition, micronutrient content, such as circulating antioxidant defences, should also be taken into consideration. Indeed, a positive relationship between oxidative stress and glycated levels of bovine serum albumin has been reported [93]. One hypothesis is that terrestrial predators have higher systemic oxidative stress levels compared to other species, which may be explained by defects in their antioxidant defences. Uric acid is one of the main non-enzymatic antioxidants in birds [94–96], and uric acid levels are especially high in carnivore species that have a rich protein diet [95, 97–100]. We should therefore take into account other important antioxidants such as vitamin E, vitamin A and carotenoids, which may be less abundant in the diets of terrestrial carnivores (e.g. [101], but see [52]). The question of whether the diet of aquatic carnivores provides a better intake of antioxidants would therefore requires a more detailed description of dietary habits. The herbivorous diet, meanwhile, despite the expected possibility contributing to lower glycaemia and glycation levels due to higher levels of PUFA compared to SFA (saturated fatty acids) (effects reviewed in [92] for humans) and higher fibre content (see [102]), did not lead to significantly lower levels of either of these two parameters. However, the relatively low sample size in that group or the presence of outliers as *Nesoenas mayeri*, with the highest glucose levels of this dataset, makes the interpretation of the obtained results somehow limited. Therefore, further research should be carried out on these species to determine if the expected pattern would emerge with a better sampling.

Finally, differences between captive and wild populations could be considered as a source of variation in glucose or glycation levels, due to a more sedentary lifestyle in captivity, with lower activity levels and a higher food intake likely to lead to increased circulating glucose levels in captive individuals. Additionally, differences in nutrient intake, such as antioxidants, specific fatty acids or amino acids, between captive and wild populations could contribute to variations in glycation levels, as we saw above. Nevertheless, we think that this factor is unlikely to significantly affect our findings regarding diet categories, because our study encompasses species from both captive and wild populations across various diet groups, particularly those exhibiting significant differences (e.g., omnivores versus terrestrial carnivores).

Conclusion and perspectives

In conclusion, the avian plasma glucose levels measured here are generally higher or in the high range of values recorded for mammalian species [28]. Our study also concludes that there is considerable variation in plasma glucose levels and albumin glycation rates between bird species, with those with the highest glucose levels showing greater resistance to glycation. The correlation between plasma glucose and life history traits are primarily influenced by its inverse association with body mass, along with non-linear, mass-independent effects of longevity. Finally, although diet does not explain plasma glucose levels in our study, terrestrial carnivores have higher albumin glycation rates than omnivores. Whether these intriguing results are explained by specific effects of certain dietary components, such as PUFAs and/or antioxidants, remains to be determined. Differences in plasma glucose levels and albumin glycation rates across bird species do not seem consistent with predictions according to the POL hypothesis, except for body mass. Further investigation is needed to elucidate the correlation between these traits and specific life conditions, such as reproductive strategy, migration patterns, flight mode or more detailed diet composition. In addition, more in-depth exploration of glycation levels within high glycaemic groups such as the Passeriformes and other small birds, which make up a significant proportion of avian species, could provide valuable new insights. Similarly, investigating groups such as the Phoenicopteriformes or Anseriformes, which are at the other end of the glycaemic-glycation spectrum, could shed light on the origin of differences between avian orders. Furthermore, notable variations were observed between species within orders, as in the Procellariiformes, which with their high mass-adjusted longevity, and particularly given that some of the age known individuals in our dataset (indeed, two specimens of snow petrels, *Pagodroma nivea*, showed ages higher than the maximum lifespan reported in the sources we explored, determining a new record) have still relatively low levels of glycation, may suggest that some species (the ones that do show higher glycation levels) are not intrinsically resistant to glycation, but rather to its adverse consequences for health. As the main limitations from this study, the usage of individuals from both wild and captive populations, sampled at different periods of the year, and the low sample size for some species, due to logistic constraints, may have introduced noise in the values reported that should be addressed in future studies by implementing a stricter sampling protocol. Also, a more thorough and accurate report and compilation of life history traits from multiple species would allow to increase the number of species included in this kind of analyses. Future research should also focus on specific species to unravel the physiological mechanisms mediating the effects of blood glucose and protein glycation on life-history trade-offs, in particular mechanisms that may vary between taxa and contribute to characteristic adaptations in birds to mitigate the potentially negative effects of comparatively high glycaemia.

Data accessibility

All data used, including references for the values taken from bibliography or databases, will be available as Electronic Supplementary Material (ESM5).

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Additional information

Ethics statement

This study followed all the legal considerations, with the ethic authorisations from the French Ministry of Secondary Education and Research, n°32475 for the zebra finches sampling, the Swiss Veterinary Office (FSVO) n° 34497 for the Alpine swifts, the Ethics Committee of the University of Extremadura (licenses112//2020 and 202//2020) and the Government of Extremadura (licenses CN0012/22/ACA and CN0063/21/ACA) for the godwits and terns from Spain, and Sampling in Terres Australes et Antarctique Françaises was approved by a Regional Animal Experimentation Ethical Committee (French Ministry of Secondary Education and Research permit APAFIS #31773-2019112519421390 v4 and APAFIS#16465–2018080111195526 v4) and by the Comité de l'Environnement Polaire and CNPN (A-2021-55). The samples from the Mulhouse zoo were taken by its licensed veterinary with capacity number: 2020-247-SPAE-162.

Authors' contributions

F. Criscuolo and F. Bertile conceived the idea, directed most of the sample collection and logistics and contributed significantly to the writing. A. Moreno-Borralló contributed to the development of the questions, gathered the diet and life-history data, performed the statistical analyses, and led the writing. S. Jaramillo Ortiz and C. Schaeffer performed the mass spectrometry analyses for protein glycation measurements, and S. Jaramillo Ortiz contributed to the glucose measures and commented on the manuscript. T. Boulinier. and V. A. Viblanc coordinated the ECONERGY and ECOPATH polar programs, organised the collection of samples on subantarctic seabirds, and commented on the manuscript. Olivier Chastel collaborated collecting marine bird samples and commented on the manuscript. B. Rey contributed with the collection of samples from Parc des Oiseaux (Villars-les-Dombes, France) and commented on the manuscript. P. Bize leads the monitoring of Alpine swifts' populations from which the samples were obtained, he helped collecting them and commented on the manuscript. J. S. Gutiérrez and J. A. Masero contributed with samples from Spain and part of the statistic scripts and commented on the manuscript. B. Quintard leads the health monitoring of Mulhouse zoo bird collection, organized and realized most of Mulhouse zoo samplings and commented on the manuscript. All authors gave their approval for publication.

Additional files

ESM 1-6 [↗](#)

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Reviewer #2 (Public review):

Summary

In this extensive comparative study, Moreno-Borralló and colleagues examine the relationships between plasma glucose levels, albumin glycation levels, diet and life-history traits across birds. Their results confirmed the expected positive relationship between plasma blood glucose level and albumin glycation rate but also provided findings that are somewhat surprising or contrast with findings of some previous studies (positive relationships between blood glucose and lifespan, or absent relationships between blood glucose and clutch mass or diet). This is the first extensive comparative analysis of glycation rates and their relationships

to plasma glucose levels and life history traits in birds that is based on data collected in a single study, with blood glucose and glycation measured using unified analytical methods (except for blood glucose data for 13 species collected from a database).

Strengths

This is an emerging topic gaining momentum in evolutionary physiology, which makes this study a timely, novel and important contribution. The study is based on a novel data set collected by the authors from 88 bird species (67 in captivity, 21 in the wild) of 22 orders, except for 13 species, for which data were collected from a database of veterinary and animal care records of zoo animals (ZIMS). This novel data set itself greatly contributes to the pool of available data on avian glycemia, as previous comparative studies either extracted data from various studies or a ZIMS database (therefore potentially containing much more noise due to different methodologies or other unstandardised factors), or only collected data from a single order, namely Passeriformes. The data further represents the first comparative avian data set on albumin glycation obtained using a unified methodology. The authors used LC-MS to determine glycation levels, which does not have problems with specificity and sensitivity that may occur with assays used in previous studies. The data analysis is thorough, and the conclusions are substantiated. Overall, this is an important study representing a substantial contribution to the emerging field evolutionary physiology focused on ecology and evolution of blood/plasma glucose levels and resistance to glycation.

Weaknesses

Unfortunately, the authors did not record handling time (i.e., time elapsed between capture and blood sampling), which may be an important source of noise because handling-stress-induced increase in blood glucose has previously been reported. Moreover, the authors themselves demonstrate that handling stress increases variance in blood glucose levels. Both effects (elevated mean and variance) are evident in Figure ESM1.2. However, this likely makes their significant findings regarding glucose levels and their associations with lifespan or glycation rate more conservative, as highlighted by the authors.

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Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

The paper explored cross-species variance in albumin glycation and blood glucose levels in the function of various life-history traits. Their results show that

- (1) blood glucose levels predict albumin glycation rates*
- (2) larger species have lower blood glucose levels*
- (3) lifespan positively correlates with blood glucose levels and*
- (4) diet predicts albumin glycation rates.*

The data presented is interesting, especially due to the relevance of glycation to the ageing process and the interesting life-history and physiological traits of birds. Most importantly, the results suggest that some mechanisms might exist that limit the level of glycation in species with the highest blood glucose levels.

While the questions raised are interesting and the amount of data the authors collected is impressive, I have some major concerns about this study:

(1) The authors combine many databases and samples of various sources. This is understandable when access to data is limited, but I expected more caution when combining these. E.g. glucose is measured in all samples without any description of how handling stress was controlled for. E.g. glucose levels can easily double in a few minutes in birds, potentially introducing variation in the data generated. The authors report no caution of this effect, or any statistical approaches aiming to check whether handling stress had an effect here, either on glucose or on glycation levels.

(2) The database with the predictors is similarly problematic. There is information pulled from captivity and wild (e.g. on lifespan) without any confirmation that the different databases are comparable or not (and here I'm not just referring to the correlation between the databases, but also to a potential systematic bias (e.g. captive-based sources likely consistently report longer lifespans). This is even more surprising, given that the authors raise the possibility of captivity effects in the discussion, and exploring this question would be extremely easy in their statistical models (a simple covariate in the MCMCglms).

(3) The authors state that the measurement of one of the primary response variables (glycation) was measured without any replicability test or reference to the replicability of the measurement technique.

(4) The methods and results are very poorly presented. For instance, new model types and variables are popping up throughout the manuscript, already reporting results, before explaining what these are e.g. results are presented on "species average models" and "model with individuals", but it's not described what these are and why we need to see both. Variables, like "centered log body mass", or "mass-adjusted lifespan" are not explained. The results section is extremely long, describing general patterns that have little relevance to the questions raised in the introduction and would be much more efficiently communicated visually or in a table.

Reviewer #2 (Public review):

Summary

In this extensive comparative study, Moreno-Borralló and colleagues examine the relationships between plasma glucose levels, albumin glycation levels, diet, and life history traits across birds. Their results confirmed the expected positive relationship between plasma blood glucose level and albumin glycation rate but also provided findings that are somewhat surprising or contradicting findings of some previous studies (relationships with lifespan, clutch mass, or diet). This is the first extensive comparative analysis of glycation rates and their relationships to plasma glucose levels and life history traits in birds that are based on data collected in a single study and measured using unified analytical methods.

Strengths

This is an emerging topic gaining momentum in evolutionary physiology, which makes this study a timely, novel, and very important contribution. The study is based on a novel data set collected by the authors from 88 bird species (67 in captivity, 21 in the wild) of 22 orders, which itself greatly contributes to the pool of available data on avian glycemia, as previous comparative studies either extracted data from various studies or a database of veterinary records of zoo animals (therefore potentially containing much more noise due to different methodologies or other unstandardised factors), or only

collected data from a single order, namely Passeriformes. The data further represents the first comparative avian data set on albumin glycation obtained using a unified methodology. The authors used LC-MS to determine glycation levels, which does not have problems with specificity and sensitivity that may occur with assays used in previous studies. The data analysis is thorough, and the conclusions are mostly wellsupported (but see my comments below). Overall, this is a very important study representing a substantial contribution to the emerging field of evolutionary physiology focused on the ecology and evolution of blood/plasma glucose levels and resistance to glycation.

Weaknesses

My main concern is about the interpretation of the coefficient of the relationship between glycation rate and plasma glucose, which reads as follows: "Given that plasma glucose is logarithm transformed and the estimated slope of their relationship is lower than one, this implies that birds with higher glucose levels have relatively lower albumin glycation rates for their glucose, fact that we would be referring as higher glycation resistance" (lines 318-321) and "the logarithmic nature of the relationship, suggests that species with higher plasma glucose levels exhibit relatively greater resistance to glycation" (lines 386-388). First, only plasma glucose (predictor) but not glycation level (response) is logarithm transformed, and this semi-logarithmic relationship assumed by the model means that an increase in glycation always slows down when blood glucose goes up, irrespective of the coefficient. The coefficient thus does not carry information that could be interpreted as higher (when <1) or lower (when >1) resistance to glycation (this only can be done in a log-log model, see below) because the semi-log relationship means that glycation increases by a constant amount (expressed by the coefficient of plasma glucose) for every tenfold increase in plasma glucose (for example, with glucose values 10 and 100, the model would predict glycation values 2 and 4 if the coefficient is 2, or 0.5 and 1 if the coefficient is 0.5). Second, the semi-logarithmic relationship could indeed be interpreted such that glycation rates are relatively lower in species with high plasma glucose levels. However, the semi-log relationship is assumed here a priori and forced to the model by log-transforming only glucose level, while not being tested against alternative models, such as: (i) a model with a simple linear relationship (glycation $\sim$ glucose); or (ii) a loglog model ($\log(\text{glycation}) \sim \log(\text{glucose})$) assuming power function relationship (glycation = $a * \text{glucose}^b$). The latter model would allow for the interpretation of the coefficient (b) as higher (when <1) or lower (when >1) resistance in glycation in species with high glucose levels as suggested by the authors.

Besides, a clear explanation of why glucose is log-transformed when included as a predictor, but not when included as a response variable, is missing.

We apologize for missing an answer to this part before. Indeed, glucose is always log transformed and this is explained in the text.

The models in the study do not control for the sampling time (i.e., time latency between capture and blood sampling), which may be an important source of noise because blood glucose increases because of stress following the capture. Although the authors claim that "this change in glucose levels with stress is mostly driven by an increase in variation instead of an increase in average values" (ESM6, line 46), their analysis of Tomasek et al.'s (2022) data set in ESM1 using Kruskal-Wallis rank sum test shows that, compared to baseline glucose levels, stress-induced glucose levels have higher median values, not only higher variation.

Although the authors calculated the variance inflation factor (VIF) for each model, it is not clear how these were interpreted and considered. In some models, $\text{GVIF}^{1/(2 * \text{Df})}$ is higher than 1.6, which indicates potentially important collinearity; see for example <https://doi.org/10.7554/eLife.103205.2>

[//www.bookdown.org/rwnahhas/RMPH/mlr-collinearity.html](https://www.bookdown.org/rwnahhas/RMPH/mlr-collinearity.html)). This is often the case for body mass or clutch mass (e.g. models of glucose or glycation based on individual measurements).

It seems that the differences between diet groups other than omnivores (the reference category in the models) were not tested and only inferred using the credible intervals from the models. However, these credible intervals relate to the comparison of each group with the reference group (Omnivore) and cannot be used for pairwise comparisons between other groups. Statistics for these contrasts should be provided instead. Based on the plot in Figure 4B, it seems possible that terrestrial carnivores differed in glycation level not only from omnivores but also from herbivores and frugivores/nectarivores.

Given that blood glucose is related to maximum lifespan, it would be interesting to also see the results of the model from Table 2 while excluding blood glucose from the predictors. This would allow for assessing if the maximum lifespan is completely independent of glycation levels. Alternatively, there might be a positive correlation mediated by blood glucose levels (based on its positive correlations with both lifespan and glycation), which would be a very interesting finding suggesting that high glycation levels do not preclude the evolution of long lifespans.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Line 84: "glycation scavengers" such as polyamines - can you specify what these polyamines do exactly?

A clarification of what we mean with "glycation scavengers" is added.

(2) Line 87-89: specify that the work of Wein et al. and this sentence is about birds.

This is now clarified.

(3) Line 95: "88 species" add "OF BIRDS". Also, I think it would be nice if you specified here that you are relying on primary data.

This is now clarified (line 96).

(4) Line 90-119: I find this paragraph very long and complex, with too many details on the methodology. For instance, I agree with listing your hypothesis, e.g. that with POL, but then what variables you use to measure the pace of life can go in the materials and methods section (so all lines between 112-119).

This is explained here as a previous reviewer considered this presentation was indeed needed in the introduction.

(5) Line 122-124: The first sentence should state that you collected blood samples from various sources, and list some examples: zoos? collaborators? designated wild captures? Stating the sample size before saying what you did to get them is a bit weird. Besides, you skipped a very important detail about how these samples were collected, when, where, and using what protocols. We know very well, that glucose levels can increase quickly with handling stress. Was this considered during the captures? Moreover, you state that you had 484 individuals, but how many samples in total? One per individual or more?

We kindly ask the reviewer to read the multiple supplementary materials provided, in which the questions of source of the samples, potential stress effects and sample sizes for each model are addressed. All individuals contributed with one sample. More details about the general sources employed are given now in lines 125-127.

| (6) Line 135-36: numbers below 10 should be spelled out.

Ok. Now that is changed.

| (7) Line 136: the first time I saw that you had both wild and captive samples. This should be among the first things to be described in the methods, as mentioned above.

As stated above, details on this are included in the supplementary materials, but further clarifications have now been included in the main text (question 5).

| (8) Line 137-138: not clear. So you had 46 samples and 9 species. But what does the 3-3-3 sample mean? or for each species you chose 9 samples (no, cause that would be 81 samples in total)?

This has now been clarified (lines 139-140).

| (9) Line 139-141: what methodological constraints? Too high glucose levels? Too little plasma?

There were cases in which the device (glucometer) produced an unspecific error. This did not correspond to too high nor too low glucose levels, as these are differently signalled errors. Neither the manual nor the client service provided useful information to discern the cause. This may perhaps be related to the composition of the plasma of certain species, interfering with the measurement. Some clarifications have been added (lines 143-146).

| (10) Line 143: should be ZIMS.

Corrected.

| (11) Line 120-148: you generally talk about individuals here, but I feel it would be more precise to use 'samples'.

The use is totally interchangeable, as we never measured more than one sample for a given individual within this study. Besides, in some cases, saying “sample” could result less informative.

| (12) Line 150: missing the final number of measurements for glucose and glycation.

Please, read the ESM6 (Table ESM6.1), where this information is given.

| (13) Line 154-155: so you took multiple samples from the same individual? It's the first time the text indicates so. Or do you mean technical replicates were not performed on the same samples?

As previously indicated, each individual included only one sample. Replicates were done only for some individuals to validate the technique, as it would be unfeasible to perform replicates of all of them. This part of the text is referring to the fact that not all samples were analysed at the same time, as it takes a considerable amount of time, and the mass spectrometry

devices are shared by other teams and project. Clarifications in this sense are now added (lines 160-163).

(14) Line 171-172: "After realizing that diet classifications from AVONET were not always suitable for our purpose" - too informal. Try rephrasing, like "After determining that AVONET diet classifications did not align with our research needs...", but you still need to specify what was wrong with it and what was changed, based on what argument?

The new formulation suggested by the reviewer has now been applied (lines 181-183). The details are given in the ESM6, as indicated in the text.

(15) Line 174-176: You start a new paragraph, talking about missing values, but you do not specify what variable are you talking about. you talk about calculating means, but the last variable you mentioned was diet, so it's even more strange.

We refer to life history traits. It has now been clarified in the text (line 185).

(16) Line 177: what longevity records? Coming from where? How did you measure longevity? Maximum lifespan ever recorded? 80-90% longevity, life expectancy???

We refer to maximum lifespan, as indicated in the introduction and in every other case throughout the manuscript. Clarifications have now been introduced (188-190).

(17) Line 180-183: using ZIMS can be problematic, especially for maximum longevity. There are often individuals who had a wrong date of birth entered or individuals that were failed to be registered as dead. The extremes in this database are often way off. If you want to combine though, you can check the correlation of lifespans obtained from different sources for the overlapping species. If it's a strong correlation it can be ok, but intuitively this is problematic.

The species for which we used ZIMS were those for which no other databases reported any values. We could try correlations for other species, but this issue is not necessarily restricted to ZIMS, as the primary origin of the data from other databases is often difficultly traceable. Also, ZIMS is potentially more updated than some of the other databases, mainly Amniotes database, from which we rely the most, as it includes the highest number of species in the most easily accessible format.

(18) Line 181-186: in ZIMS you calculate the average of the competing records, otherwise you choose the max. Why use different preferences for the same data?

This constitutes a misunderstanding, for which we include clarifications now (line 196). We were referring here to the fact that for maximum lifespan the maximum is always chosen, while for other variables an average is calculated.

(19) Line 198: Burn-in and thinning interval is quite low compared to your number of iterations. How were model convergences checked?

Please, check ESM1.

(20) Line 201-203: What's the argument using these priors? Why not use noninformative ones? Do you have some a priori expectations? If so, it should be explained.

Models have now been rerun with no expectations on the variance partitions so the priors are less informative, given the lack of firm expectations, and results are similar. Smaller nu

values are also tried.

| (21) Line 217: "carried" OUT.

Corrected (now in line 229).

| (22) Line 233-234: "species average model" - what is this? it was not described in the methods.

Please, read the ESM6.

| (23) Line 232-246: (a) all this would be better described by a table or plot. You can highlight some interesting patterns, but describing it all in the text is not very useful I think, (b) statistically comparing orders represented by a single species is a bit odd.

(a) Figure 1 shows this graphically, but this part was found to be quite short without descriptions by previous reviewers. (b) We recognise this limitation, but this part is not presented as one of the main results of the article, and just constitutes an attempt to illustrate very general patterns, in order to guide future research, as in most groups glycation has never been measured, so this still constitutes the best illustration of such patterns in the literature.

| (24) Line 281: the first time I saw "mass-adjusted maximum lifespan" - what is this, and how was it calculated? It should be described in the methods. But in any case, neither ratios, nor residuals should be used, but preferably the two variables should be entered side by side in the model.

Please, see ESM6 for the explanations and justifications for all of this.

| (25) Line 281: there was also no mention of quadratic terms so far. How were polynomial effects tested/introduced in the models? Orthogonal polynomials? or $x + x^2$?

Please, read ESM6.

| (26) Table 1. What is 'Centred Log10Body mass', should be added in the methods.

Please, read ESM6.

| (27) Table 1: what's the argument behind separating terrestrial and aquatic carnivores?

This was mostly based on the a priori separation made in AVONET, but it is also used in a similar way by Szarka and Lendvai 2024 (comparative study on glucose in birds), where differences in glucose levels between piscivorous and carnivorous are reported. We had some reasons to think that certain differences in dietary nutrient composition, as discussed later, can make this difference relevant.

| (28) Table 1: The variable "Maximum lifespan" is discussed and plotted as 'massadjusted maximum lifespan' and 'residual maximum lifespan'. First, this is confusing, the same name should be used throughout and it should be defined in the methods section. Second, it seems that non-linear effects were tested by using $x + x^2$. This is problematic statistically, orthogonal polynomials should be used instead (check polyfunction in R). Also, how did you decide to test for non-linear effects in the case of lifespan but not the other continuous predictors? Should be described in the methods again.

Please, read ESM6. Data exploration was performed prior to carry out these models. Orthogonal polynomials were considered to difficult the interpretation of the estimates and therefore the patterns predicted by the models, so raw polynomials were used. Clarifications have now been included in line 297.

(29) Figure 2. From the figure label, now I see that relative lifespan is in fact residual. This is problematic, see Freckleton, R. P. (2009). The seven deadly sins of comparative analysis. *Journal of evolutionary biology*, 22(7), 1367-1375. Using body mass and lifespan side by side is preferred. This would also avoid forcing more emphasis on body mass over lifespan meaning that you subjectively introduce body mass as a key predictor, but lifespan and body size are highly correlated, so by this, you remove a large portion of variance that might in fact be better explained by lifespan.

Please, read ESM6 for justifications on the use of residuals.

Reviewer #2 (Recommendations for the authors):

(1) If the semi-logarithmic relationship (glycation $\sim \log_{10}(\text{glucose})$) is to be used to support the hypothesis about higher glycation resistance in species with high blood glucose (lines 318-321 and 386-388), it should be tested whether it is significantly better than the model assuming a simple linear relationship (i.e., glycation $\sim \text{glucose}$). Alternatively, if the coefficient is to be used to determine whether glycation rate slows down or accelerates with increasing glucose levels, log-log model ($\log_{10}(\text{glycation}) \sim \log_{10}(\text{glucose})$) assuming power function relationship (glycation = $a * \text{glucose}^b$) should be used (as is for example in the literature about relationships between metabolic rates and body size). Probably the best approach would be to compare all three models (linear, semi-logarithmic, and log-log) and test if one performs significantly better. If none of them, then the linear model should be selected as the most parsimonious.

Different options (linear, both semi-logarithmic combinations and log-log) have now been tested, with similar results. All of the models confirm the pattern of a significant positive relationship between glucose and glycation. Moreover, when standardizing the variables (both glucose and glycation, either log transformed or not), the estimate of the slope is almost equal for all the models. It is also lower than one, which in the case of both the linear and log-log confirms the stated prediction. The log-log model, showing a much lower DIC than the linear version, is now shown as the final model.

(2) ESM6, line 46: Please note that Kruskal-Wallis rank sum test in ESM1 shows that, compared to baseline glucose levels, stress-induced glucose levels have higher median values (not only higher variation). With this in mind, what is the argument here about increased variation being the main driver of stress-induced change in glucose levels based on? It seems that both the median values and variation differ between baseline and stress-induced levels, and this should be acknowledged here.

As discussed in the public answers, Kruskal Wallis does not allow to determine differences in mean, but just says that the groups are “different” (implicitly, in their ranksums, which does not mean necessarily in mean), while the Levene test performed signals heteroskedasticity. This makes this feature of the data analytically more grounded. Of course, when looking at the data, a higher mean can be perceived, but nothing can be said about its statistical significance. Still, some subtle changes have been introduced in corresponding section of the ESM6.

(3) Have you recorded the sampling times? If yes, why not control them in the models? It is at least highly advisable to include the sampling times in the data (ESM5).

As indicated in ESM6 lines 42-43, we do not have sampling times for most of the individuals (only zebra finches and swifts), so this cannot be accounted for in the models.

(4) If sampling times will remain uncontrolled statistically, I recommend mentioning this fact and its potential consequences (i.e., rather conservative results) in the Methods section of the main text, not only in ESM6.

A brief description of this has now been included in the main text (lines 129-132), referencing the more detailed discussion on the supplementary materials. Some subtle changes have also been included in the “Possible effects of stress” section of the ESM6.

(5) ESM6, lines 52-53: The lower repeatability in Tomasek et al.' study compared to your study is irrelevant to the argument about the conservative nature of your results (the difference in repeatability between both studies is most probably due to the broader taxonomic coverage of the current study). The important result in this context is that repeatability is lower when sampling time is not considered within Tomasek et al's data set (ESM1). Therefore, I suggest rewording "showing a lower species repeatability than that from our data" to "showing lower species repeatability when sampling time is not considered" to avoid confusion. Please also note that you refer here to species repeatability but, in ESM1, you calculate individual repeatability. Nevertheless, both individual and species repeatabilities are lower when not controlling for sampling time because the main driver, in that case, is an increased residual variance.

We recognize the current confusion in the way the explanation is exposed, and have significantly changed the redaction of the section. However, we would like to indicate that ESM1 shows both species and individual repeatability (for Tomasek et al. 2022 data, for ours only species as we do not have repeated individual values). Changes are now made to make it more evident.

(6) I recommend providing brief guidelines for the interpretation of VIFs to the readers, as well as a brief discussion of the obtained values and their potential importance.

Thank you for the recommendation. We included a brief description in lines 230-231. Also in the results section (lines 389-393).

(7) Line: 264: Please note that the variance explained by phylogeny obtained from the models with other (fixed) predictors does not relate to the traits (glucose or glycation) per se but to model residuals.

We appreciate the indication, and this has been rephrased accordingly (lines 280-286).

(8) Change the term "confidence intervals" to "credible intervals" throughout the paper, since confidence interval is a frequentist term and its interpretations are different from Bayesian credible interval.

Thank you for the remark, this has now been changed.

(9) Besides lifespan, have you also considered quadratic terms for body mass? The plot in Figure 2A suggests there might be a non-linear relationship too.

A quadratic component of body mass has not shown any significant effect on glucose in an alternative model. Also, a model with linear instead of log glucose (as performed in other studies) did not perform better by comparing the DICs, despite both showing a significant

relationship between glucose and body mass. Therefore, this model remains the best option considered as presented in the manuscript.

(10) ESM6, lines 115-116: *It is usually recommended that only factors with at least 6 or 8 levels are included as random effects because a lower number of levels is insufficient for a good estimation of variance.*

In a Bayesian approach this does not apply, as random and fixed factors are estimated similarly.

(11) *Typos and other minor issues:*

a) Line 66: Delete "related".

b) Figure 2: "B" label is missing in the plot.

c) Reference 9: Delete "Author".

d) References 15 and 83 are duplicated. Keep only ref. 83, which has the correct citation details.

e) ESM6, line 49: Change "GLLM" to "GLMM".

Thank you for indicating this. Now it's corrected.

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